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1. The functions x and y satisfy

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \quad y = 5x$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} + 2y = 6x$$

(a) Differentiate the first equation and eliminate y to show that x satisfies

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} - 4x = 0$$ **[4]**

(b) Given that $x = 1$ and $y = 3$ when $t = 0$, find explicit expressions for x and y in terms of t. **[7]**

Solution

(a) From

$$\frac{\mathrm{d}x}{\mathrm{d}t} + y = 5x$$

we first rearrange to express y in terms of x:

$$y = 5x - \frac{\mathrm{d}x}{\mathrm{d}t}$$

Differentiate the first equation with respect to t:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}y}{\mathrm{d}t} = 5\frac{\mathrm{d}x}{\mathrm{d}t}$$

From the second equation,

$$\frac{\mathrm{d}y}{\mathrm{d}t} + 2y = 6x$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 6x - 2y$$

Substitute this into the differentiated equation:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6x - 2y = 5\frac{\mathrm{d}x}{\mathrm{d}t}$$

Now substitute $y = 5x - \frac{\mathrm{d}x}{\mathrm{d}t}$:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6x - 2\left(5x - \frac{\mathrm{d}x}{\mathrm{d}t}\right) = 5\frac{\mathrm{d}x}{\mathrm{d}t}$$
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6x - 10x + 2\frac{\mathrm{d}x}{\mathrm{d}t} = 5\frac{\mathrm{d}x}{\mathrm{d}t}$$
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4x + 2\frac{\mathrm{d}x}{\mathrm{d}t} - 5\frac{\mathrm{d}x}{\mathrm{d}t} = 0$$
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} - 4x = 0$$

Hence,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} - 4x = 0$$

(b) We solve the differential equation found in part (a):

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 3\frac{\mathrm{d}x}{\mathrm{d}t} - 4x = 0$$

The auxiliary equation is

$$\lambda^2 - 3\lambda - 4 = 0$$

Factorising,

$$(\lambda - 4)(\lambda + 1) = 0$$

So the roots are $\lambda = 4$ and $\lambda = -1$, giving

$$x = A\mathrm{e}^{4t} + B\mathrm{e}^{-t}$$

Differentiate:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 4A\mathrm{e}^{4t} - B\mathrm{e}^{-t}$$

From the first original equation,

$$\frac{\mathrm{d}x}{\mathrm{d}t} + y = 5x$$

so

$$y = 5x - \frac{\mathrm{d}x}{\mathrm{d}t}$$

Substitute the expressions for x and $\frac{\mathrm{d}x}{\mathrm{d}t}$:

$$\begin{aligned} y &= 5\left(A\mathrm{e}^{4t} + B\mathrm{e}^{-t}\right) - \left(4A\mathrm{e}^{4t} - B\mathrm{e}^{-t}\right) \\ &= 5A\mathrm{e}^{4t} + 5B\mathrm{e}^{-t} - 4A\mathrm{e}^{4t} + B\mathrm{e}^{-t} \\ &= A\mathrm{e}^{4t} + 6B\mathrm{e}^{-t} \end{aligned}$$

Now use the initial conditions $x = 1$ and $y = 3$ when $t = 0$.
From $x(0) = 1$:

$$A + B = 1$$

From $y(0) = 3$:

$$A + 6B = 3$$

Subtract the first equation from the second:

$$\begin{aligned} (A + 6B) - (A + B) &= 3 - 1 \\ 5B &= 2 \\ B &= \frac{2}{5} \end{aligned}$$

Then

$$\begin{aligned} A + \frac{2}{5} &= 1 \\ A &= \frac{3}{5} \end{aligned}$$

Substitute these values into x and y:

$$x = \frac{3}{5}\mathrm{e}^{4t} + \frac{2}{5}\mathrm{e}^{-t}$$

and

$$y = \frac{3}{5}\mathrm{e}^{4t} + \frac{12}{5}\mathrm{e}^{-t}$$

Therefore the explicit expressions are

$$x = \frac{3}{5}\mathrm{e}^{4t} + \frac{2}{5}\mathrm{e}^{-t}, \qquad y = \frac{3}{5}\mathrm{e}^{4t} + \frac{12}{5}\mathrm{e}^{-t}$$

2. The functions x and y satisfy the simultaneous differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} + y = 0$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} - x = 2e^t$$

and when $t = 0$, $x = 0$ and $y = 1$.

Solve the system, giving x and y explicitly in terms of t **[11]**

Solution

From the first differential equation,

$$\frac{\mathrm{d}x}{\mathrm{d}t} + y = 0$$

we have

$$y = -\frac{\mathrm{d}x}{\mathrm{d}t}$$

Differentiate this with respect to t:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

Now substitute y and $\frac{\mathrm{d}y}{\mathrm{d}t}$ into

$$\frac{\mathrm{d}y}{\mathrm{d}t} - x = 2e^t$$

This gives

$$-\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - x = 2e^t$$
$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x = -2e^t$$

So we solve the second order linear equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x = -2e^t$$

We also need initial conditions for x.
We are given

$$x(0) = 0$$

Also, when $t = 0$, the first equation gives

$$\frac{\mathrm{d}x}{\mathrm{d}t} + y = 0$$
$$\frac{\mathrm{d}x}{\mathrm{d}t} + 1 = 0$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}t}(0) = -1$$

Now solve for x.
For the complementary function x_c, the auxiliary equation is

$$\lambda^2 + 1 = 0$$

so the roots are $\lambda = \pm i$.
Hence

$$x_c = A\cos t + B\sin t$$

For a particular integral x_p, since the right-hand side is $-2e^t$, try

$$x_p = Ce^t$$

Then

$$\frac{\mathrm{d}^2 x_p}{\mathrm{d}t^2} = Ce^t$$

Substituting into the differential equation:

$$\begin{aligned} Ce^t + Ce^t &= -2e^t \\ 2Ce^t &= -2e^t \end{aligned}$$

so

$$C = -1$$

Thus

$$x_p = -e^t$$

Therefore the general solution for x is

$$x = A\cos t + B\sin t - e^t$$

Use the initial conditions.
First, $x(0) = 0$:

$$\begin{aligned} A\cos 0 + B\sin 0 - e^0 &= 0 \\ A - 1 &= 0 \end{aligned}$$

so

$$A = 1$$

Now differentiate x:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -A\sin t + B\cos t - e^t$$

Use $\frac{\mathrm{d}x}{\mathrm{d}t}(0) = -1$:

$$\begin{aligned} -A\sin 0 + B\cos 0 - e^0 &= -1 \\ B - 1 &= -1 \end{aligned}$$

so

$$B = 0$$

Hence

$$x = \cos t - e^t$$

Now find y from

$$y = -\frac{\mathrm{d}x}{\mathrm{d}t}$$

Since

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t - e^t$$

it follows that

$$y = \sin t + e^t$$

So the required explicit solutions are

$$x = \cos t - e^t, \qquad y = \sin t + e^t$$

3. The functions x and y satisfy

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 3x + 4y + 7$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -x - y - 1$$

Given that $x = 5$ and $y = -3$ when $t = 0$, find x and y as functions of t **[10]**

Solution

From the second differential equation,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -x - y - 1$$

so

$$x = -\frac{\mathrm{d}y}{\mathrm{d}t} - y - 1$$

Differentiate this to obtain an expression for $\frac{\mathrm{d}x}{\mathrm{d}t}$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}$$

Now substitute both expressions into

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 3x + 4y + 7$$

This gives

$$\begin{aligned} -\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t} &= 3\left(-\frac{\mathrm{d}y}{\mathrm{d}t} - y - 1\right) + 4y + 7 \\ &= -3\frac{\mathrm{d}y}{\mathrm{d}t} - 3y - 3 + 4y + 7 \\ &= -3\frac{\mathrm{d}y}{\mathrm{d}t} + y + 4 \end{aligned}$$

Rearranging,

$$\begin{aligned} -\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t} &= -3\frac{\mathrm{d}y}{\mathrm{d}t} + y + 4 \\ -\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} - y &= 4 \\ \frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\frac{\mathrm{d}y}{\mathrm{d}t} + y &= -4 \end{aligned}$$

We now solve this second-order linear differential equation.
The complementary function y_c satisfies the auxiliary equation

$$\lambda^2 - 2\lambda + 1 = 0$$

so

$$(\lambda - 1)^2 = 0$$

Hence

$$y_c = (A + Bt)e^t$$

For a particular integral y_p, try a constant $y_p = k$. Substituting into

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\frac{\mathrm{d}y}{\mathrm{d}t} + y = -4$$

gives

$$0 - 0 + k = -4$$

so

$$k = -4$$

Therefore,

$$y = (A + Bt)e^t - 4$$

Use the initial condition $y(0) = -3$:

$$\begin{aligned}(A + 0)e^0 - 4 &= -3\\ A - 4 &= -3\\ A &= 1\end{aligned}$$

So

$$y = (1 + Bt)e^t - 4$$

Now find x using

$$x = -\frac{\mathrm{d}y}{\mathrm{d}t} - y - 1$$

First differentiate y:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\big((A + Bt)e^t - 4\big)\\ &= (A + Bt)e^t + Be^t\\ &= (A + Bt + B)e^t\end{aligned}$$

So

$$\begin{aligned}x &= -(A + Bt + B)e^t - \big((A + Bt)e^t - 4\big) - 1\\ &= -(2A + 2Bt + B)e^t + 3\end{aligned}$$

Now substitute $A = 1$:

$$x = -(2 + 2Bt + B)e^t + 3$$

Use the initial condition $x(0) = 5$:

$$\begin{aligned}-(2 + B)e^0 + 3 &= 5\\ -(2 + B) + 3 &= 5\\ 1 - B &= 5\\ B &= -4\end{aligned}$$

Substitute $B = -4$ into the expressions for x and y:

$$\begin{aligned}y(t) &= (1 - 4t)e^t - 4\\ x(t) &= -(2 - 8t - 4)e^t + 3\\ &= (2 + 8t)e^t + 3\end{aligned}$$

Therefore, the required functions are $x(t) = (2 + 8t)e^t + 3$ and $y(t) = (1 - 4t)e^t - 4$.

4. The functions x and y satisfy the system of differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2y - 7$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2x + y - 8$$

(a) Find the stationary point of the system.
That is, find the values of x and y such that $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$ **[2]**

(b) Given that $x = 4$ and $y = 0$ when $t = 0$, find the particular solutions for x and y in terms of t **[8]**

Solution

(a) At a stationary point,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 0 \qquad \text{and} \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 0$$

So

$$x + 2y - 7 = 0$$
$$2x + y - 8 = 0$$

Hence

$$x + 2y = 7, \qquad 2x + y = 8$$

From the first equation,

$$x = 7 - 2y$$

Substituting into the second equation:

$$\begin{aligned} 2(7 - 2y) + y &= 8 \\ 14 - 4y + y &= 8 \\ 14 - 3y &= 8 \\ 3y &= 6 \\ y &= 2 \end{aligned}$$

Then

$$x = 7 - 2(2) = 3$$

So the stationary point is $(3, 2)$.

(b) Eliminate y directly from the system.
From

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2y - 7$$

we have

$$2y = \frac{\mathrm{d}x}{\mathrm{d}t} - x + 7$$

Differentiate

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2y - 7$$

to obtain

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{\mathrm{d}x}{\mathrm{d}t} + 2\frac{\mathrm{d}y}{\mathrm{d}t}$$

Now use

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2x + y - 8$$

so that

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= \frac{\mathrm{d}x}{\mathrm{d}t} + 2(2x + y - 8)\\ &= \frac{\mathrm{d}x}{\mathrm{d}t} + 4x + 2y - 16\end{aligned}$$

Substitute

$$2y = \frac{\mathrm{d}x}{\mathrm{d}t} - x + 7$$

to get

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= \frac{\mathrm{d}x}{\mathrm{d}t} + 4x + \left(\frac{\mathrm{d}x}{\mathrm{d}t} - x + 7\right) - 16\\ &= 2\frac{\mathrm{d}x}{\mathrm{d}t} + 3x - 9\end{aligned}$$

Therefore

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 3x = -9$$

The auxiliary equation is

$$\lambda^2 - 2\lambda - 3 = 0$$

Factorising:

$$(\lambda - 3)(\lambda + 1) = 0$$

So the complementary function is

$$x_c = A\mathrm{e}^{3t} + B\mathrm{e}^{-t}$$

For a particular integral, try a constant value

$$x_p = k$$

Substituting into

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} - 3x = -9$$

gives

$$-3k = -9$$

so

$$k = 3$$

Hence

$$x = x_c + x_p = A\mathrm{e}^{3t} + B\mathrm{e}^{-t} + 3$$

Now use the initial conditions.
Since $x(0) = 4$,

$$A + B + 3 = 4$$

so

$$A + B = 1$$

Also

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x + 2y - 7$$

and when $t = 0$,

$$\left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} = 4 + 2(0) - 7 = -3$$

But

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 3A\mathrm{e}^{3t} - B\mathrm{e}^{-t}$$

so at $t = 0$,

$$3A - B = -3$$

Solving

$$A + B = 1, \qquad 3A - B = -3$$

gives

$$A = -\frac{1}{2}, \qquad B = \frac{3}{2}$$

Therefore

$$x = 3 - \frac{1}{2}\mathrm{e}^{3t} + \frac{3}{2}\mathrm{e}^{-t}$$

Now find y from

$$2y = \frac{\mathrm{d}x}{\mathrm{d}t} - x + 7$$

Since

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{3}{2}\mathrm{e}^{3t} - \frac{3}{2}\mathrm{e}^{-t}$$

we get

$$\begin{aligned} 2y &= \left(-\frac{3}{2}\mathrm{e}^{3t} - \frac{3}{2}\mathrm{e}^{-t}\right) - \left(3 - \frac{1}{2}\mathrm{e}^{3t} + \frac{3}{2}\mathrm{e}^{-t}\right) + 7 \\ &= -\mathrm{e}^{3t} - 3\mathrm{e}^{-t} + 4 \end{aligned}$$

Hence

$$y = 2 - \frac{1}{2}\mathrm{e}^{3t} - \frac{3}{2}\mathrm{e}^{-t}$$

Therefore the particular solutions are

$$x = 3 - \frac{1}{2}\mathrm{e}^{3t} + \frac{3}{2}\mathrm{e}^{-t}, \qquad y = 2 - \frac{1}{2}\mathrm{e}^{3t} - \frac{3}{2}\mathrm{e}^{-t}$$

5. For $t \geq 0$, the functions $x = f(t)$ and $y = g(t)$ satisfy the coupled first-order differential equations.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x - y$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = x + 2y$$

Given that $x = 1$ and $y = 2$ when $t = 0$, determine exact simplified expressions for $f(t)$ and $g(t)$ **[10]**

Solution

Starting with

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x - y$$

we first express y in terms of x and $\frac{\mathrm{d}x}{\mathrm{d}t}$:

$$y = 2x - \frac{\mathrm{d}x}{\mathrm{d}t}$$

Differentiate this with respect to t:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

Now use the second differential equation:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t} &= x + 2y \\ &= x + 2\left(2x - \frac{\mathrm{d}x}{\mathrm{d}t}\right) \\ &= x + 4x - 2\frac{\mathrm{d}x}{\mathrm{d}t} \\ &= 5x - 2\frac{\mathrm{d}x}{\mathrm{d}t}\end{aligned}$$

Equating the two expressions for $\frac{\mathrm{d}y}{\mathrm{d}t}$ gives

$$\begin{aligned}2\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= 5x - 2\frac{\mathrm{d}x}{\mathrm{d}t} \\ -\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} - 5x &= 0 \\ \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x &= 0\end{aligned}$$

So we solve the second-order linear differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$$

Its auxiliary equation is

$$\lambda^2 - 4\lambda + 5 = 0$$

Solving,

$$\lambda = \frac{4 \pm \sqrt{16 - 20}}{2} = 2 \pm i$$

Hence the complementary function is

$$x = e^{2t}(A\cos t + B\sin t)$$

Use the initial condition $x(0) = 1$:

$$\begin{aligned}x(0) &= e^0(A\cos 0 + B\sin 0) \\ &= A\end{aligned}$$

so

$$A = 1$$

To find B, first find $\frac{\mathrm{d}x}{\mathrm{d}t}$ at $t = 0$ from the original equation:

$$\begin{aligned}\left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} &= 2x(0) - y(0)\\ &= 2(1) - 2\\ &= 0\end{aligned}$$

Now differentiate

$$x = e^{2t}(A\cos t + B\sin t)$$

Using the product rule,

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= 2e^{2t}(A\cos t + B\sin t) + e^{2t}(-A\sin t + B\cos t)\\ &= e^{2t}\big((2A + B)\cos t + (2B - A)\sin t\big)\end{aligned}$$

At $t = 0$,

$$\begin{aligned}0 = \left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} &= e^{0}\big((2A + B)\cos 0 + (2B - A)\sin 0\big)\\ &= 2A + B\end{aligned}$$

Since $A = 1$,

$$2(1) + B = 0$$

so

$$B = -2$$

Therefore

$$x = e^{2t}(\cos t - 2\sin t)$$

Now find y using

$$y = 2x - \frac{\mathrm{d}x}{\mathrm{d}t}$$

First differentiate $x = e^{2t}(\cos t - 2\sin t)$:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= 2e^{2t}(\cos t - 2\sin t) + e^{2t}(-\sin t - 2\cos t)\\ &= e^{2t}(2\cos t - 4\sin t - \sin t - 2\cos t)\\ &= -5e^{2t}\sin t\end{aligned}$$

So

$$\begin{aligned}y &= 2e^{2t}(\cos t - 2\sin t) - \big(-5e^{2t}\sin t\big)\\ &= 2e^{2t}\cos t - 4e^{2t}\sin t + 5e^{2t}\sin t\\ &= e^{2t}(2\cos t + \sin t)\end{aligned}$$

Hence the required functions are

$$f(t) = e^{2t}(\cos t - 2\sin t), \qquad g(t) = e^{2t}(2\cos t + \sin t)$$

6. An ecologist is studying the effect of introducing a population of weed-eating snails into a lake containing an invasive waterweed. During the first few months, a temporary pollutant also affects the snail population.

At time t months, the amount of waterweed, w, and the number of snails, s, are modelled by the differential equations

$$\frac{\mathrm{d}w}{\mathrm{d}t} = 4w - s$$
$$\frac{\mathrm{d}s}{\mathrm{d}t} = 6w - s - 48e^{-t}$$

(a) Show that

$$\frac{\mathrm{d}^2w}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w}{\mathrm{d}t} + 2w = 48e^{-t}$$ **[3]**

(b) Find a general solution for the amount of waterweed at time t months **[6]**

(c) Find a general solution for the number of snails at time t months **[2]**

The model predicts that, at time T months, the waterweed will have been eliminated.
Given that $w = 30$ and $s = 110$ when $t = 0$

(d) Find the value of T, giving your answer to 3 decimal places **[6]**

(e) Suggest one limitation of the model **[1]**

Solution

(a) From

$$\frac{\mathrm{d}w}{\mathrm{d}t} = 4w - s$$

we have

$$s = 4w - \frac{\mathrm{d}w}{\mathrm{d}t}$$

Differentiate $\frac{\mathrm{d}w}{\mathrm{d}t} = 4w - s$ with respect to t:

$$\frac{\mathrm{d}^2w}{\mathrm{d}t^2} = 4\frac{\mathrm{d}w}{\mathrm{d}t} - \frac{\mathrm{d}s}{\mathrm{d}t}$$

Now use

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 6w - s - 48e^{-t}$$

and substitute $s = 4w - \frac{\mathrm{d}w}{\mathrm{d}t}$:

$$\begin{aligned}\frac{\mathrm{d}s}{\mathrm{d}t} &= 6w - \left(4w - \frac{\mathrm{d}w}{\mathrm{d}t}\right) - 48e^{-t}\\ &= 2w + \frac{\mathrm{d}w}{\mathrm{d}t} - 48e^{-t}\end{aligned}$$

So

$$\begin{aligned}\frac{\mathrm{d}^2w}{\mathrm{d}t^2} &= 4\frac{\mathrm{d}w}{\mathrm{d}t} - \left(2w + \frac{\mathrm{d}w}{\mathrm{d}t} - 48e^{-t}\right)\\ &= 3\frac{\mathrm{d}w}{\mathrm{d}t} - 2w + 48e^{-t}\end{aligned}$$

Hence

$$\frac{\mathrm{d}^2 w}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w}{\mathrm{d}t} + 2w = 48e^{-t}$$

This is the required result.

(b) We solve

$$\frac{\mathrm{d}^2 w}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w}{\mathrm{d}t} + 2w = 48e^{-t}$$

First find the complementary function w_c from

$$\frac{\mathrm{d}^2 w}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w}{\mathrm{d}t} + 2w = 0$$

The auxiliary equation is

$$\lambda^2 - 3\lambda + 2 = 0$$

Factorising,

$$(\lambda - 1)(\lambda - 2) = 0$$

So the roots are $\lambda = 1$ and $\lambda = 2$, giving

$$w_c = Ae^t + Be^{2t}$$

For a particular integral w_p, since the right-hand side is $48e^{-t}$, try

$$w_p = ke^{-t}$$

Then

$$\frac{\mathrm{d}w_p}{\mathrm{d}t} = -ke^{-t}, \qquad \frac{\mathrm{d}^2 w_p}{\mathrm{d}t^2} = ke^{-t}$$

Substitute into the differential equation:

$$\begin{aligned}\frac{\mathrm{d}^2 w_p}{\mathrm{d}t^2} - 3\frac{\mathrm{d}w_p}{\mathrm{d}t} + 2w_p &= ke^{-t} - 3(-ke^{-t}) + 2ke^{-t} \\ &= 6ke^{-t}\end{aligned}$$

Comparing with $48e^{-t}$,

$$6k = 48$$

so

$$k = 8$$

Therefore

$$w_p = 8e^{-t}$$

Hence the general solution is

$$w = w_c + w_p = Ae^t + Be^{2t} + 8e^{-t}$$

So the general solution for the waterweed is

$$w = Ae^t + Be^{2t} + 8e^{-t}$$

(c) From part (a),

$$s = 4w - \frac{\mathrm{d}w}{\mathrm{d}t}$$

Using

$$w = Ae^t + Be^{2t} + 8e^{-t}$$

we get

$$\frac{\mathrm{d}w}{\mathrm{d}t} = Ae^t + 2Be^{2t} - 8e^{-t}$$

So

$$\begin{aligned} s &= 4\left(Ae^t + Be^{2t} + 8e^{-t}\right) - \left(Ae^t + 2Be^{2t} - 8e^{-t}\right) \\ &= 3Ae^t + 2Be^{2t} + 40e^{-t} \end{aligned}$$

Therefore the general solution for the number of snails is

$$s = 3Ae^t + 2Be^{2t} + 40e^{-t}$$

(d) Use the initial conditions $w = 30$ and $s = 110$ when $t = 0$.

From

$$w = Ae^t + Be^{2t} + 8e^{-t}$$

when $t = 0$,

$$30 = A + B + 8$$

so

$$A + B = 22$$

From

$$s = 3Ae^t + 2Be^{2t} + 40e^{-t}$$

when $t = 0$,

$$110 = 3A + 2B + 40$$

so

$$3A + 2B = 70$$

Solve the simultaneous equations:

$$\begin{aligned} A + B &= 22 \\ 3A + 2B &= 70 \end{aligned}$$

Subtract $2(A + B = 22)$ from $3A + 2B = 70$:

$$A = 26$$

Then

$$B = 22 - 26 = -4$$

So

$$w = 26e^t - 4e^{2t} + 8e^{-t}$$

The waterweed is eliminated when $w = 0$, so set $w(T) = 0$:

$$26e^T - 4e^{2T} + 8e^{-T} = 0$$

Multiply through by e^T:

$$26e^{2T} - 4e^{3T} + 8 = 0$$

Let

$$x = e^T$$

where $x > 0$. Then

$$26x^2 - 4x^3 + 8 = 0$$

Rearranging,

$$2x^3 - 13x^2 - 4 = 0$$

Solving this cubic numerically gives the positive root

$$x \approx 6.547$$

Therefore

$$T = \ln(6.547) \approx 1.879$$

So the value of T is

$$T \approx 1.879 \text{ months}$$

(e) One limitation is that the model can predict negative values of w or s, which is not realistic for populations.

So a valid limitation is: the model can give impossible negative waterweed or snail populations.

7. At the start of 2021, a survey began of the number of mature trees and saplings in a managed woodland.

At time t years after the survey began, the number of mature trees, m, and the number of saplings, s, are modelled by the differential equations

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{1}{4}s$$
$$\frac{\mathrm{d}s}{\mathrm{d}t} = -\frac{1}{20}m - \frac{1}{10}s$$

(a) Show that

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{80}s = 0$$ **[3]**

(b) Find a general solution for the number of saplings in the woodland at time t years **[4]**

(c) Hence find a general solution for the number of mature trees in the woodland at time t years **[3]**

At the start of 2021 there were 20 mature trees and 20 saplings.

(d) (i) According to this model, find the first value of t for which no saplings remain, and state the year.

(ii) According to this model, how many mature trees are there at that time?

(iii) Use your answers to parts (i) and (ii) to comment on the suitability of the model. **[7]**

Solution

(a) Differentiate the equation for $\frac{\mathrm{d}s}{\mathrm{d}t}$ with respect to t:

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{1}{20}\frac{\mathrm{d}m}{\mathrm{d}t} - \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t}$$

Using

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{1}{4}s$$

gives

$$\begin{aligned}\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} &= -\frac{1}{20}\left(\frac{1}{4}s\right) - \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t} \\ &= -\frac{1}{80}s - \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t}\end{aligned}$$

Rearranging,

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{80}s = 0$$

Hence the required equation is shown.

(b) From part (a), we solve

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} + \frac{1}{10}\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{80}s = 0$$

The auxiliary equation is

$$\lambda^2 + \frac{1}{10}\lambda + \frac{1}{80} = 0$$

Multiplying through by 80,

$$80\lambda^2 + 8\lambda + 1 = 0$$

Using the quadratic formula,

$$\begin{aligned}\lambda &= \frac{-8 \pm \sqrt{8^2 - 4(80)(1)}}{160} \\ &= \frac{-8 \pm \sqrt{64 - 320}}{160} \\ &= \frac{-8 \pm \sqrt{-256}}{160} \\ &= \frac{-8 \pm 16\mathrm{i}}{160} \\ &= -\frac{1}{20} \pm \frac{1}{10}\mathrm{i}\end{aligned}$$

Therefore the general solution is

$$s = e^{-t/20}\left(A\cos\frac{t}{10} + B\sin\frac{t}{10}\right)$$

So a general solution for the number of saplings is

$$s = e^{-t/20}\left(A\cos\frac{t}{10} + B\sin\frac{t}{10}\right)$$

(c) Use

$$\frac{\mathrm{d}s}{\mathrm{d}t} = -\frac{1}{20}m - \frac{1}{10}s$$

Rearranging for m,

$$m = -20\left(\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{10}s\right)$$

Let $\theta = \dfrac{t}{10}$. From part (b),

$$s = e^{-t/20}(A\cos\theta + B\sin\theta)$$

Differentiate:

$$\begin{aligned}\frac{\mathrm{d}s}{\mathrm{d}t} &= -\frac{1}{20}e^{-t/20}(A\cos\theta + B\sin\theta) + e^{-t/20}\left(-\frac{A}{10}\sin\theta + \frac{B}{10}\cos\theta\right) \\ &= e^{-t/20}\left[\left(\frac{B}{10} - \frac{A}{20}\right)\cos\theta + \left(-\frac{A}{10} - \frac{B}{20}\right)\sin\theta\right]\end{aligned}$$

Now substitute into the expression for m:

$$\begin{aligned}m &= -20\left(\frac{\mathrm{d}s}{\mathrm{d}t} + \frac{1}{10}s\right) \\ &= -20e^{-t/20}\left[\left(\frac{B}{10} - \frac{A}{20} + \frac{A}{10}\right)\cos\theta + \left(-\frac{A}{10} - \frac{B}{20} + \frac{B}{10}\right)\sin\theta\right] \\ &= -20e^{-t/20}\left[\left(\frac{A+2B}{20}\right)\cos\theta + \left(\frac{-2A+B}{20}\right)\sin\theta\right] \\ &= e^{-t/20}\left[(-A-2B)\cos\theta + (2A-B)\sin\theta\right]\end{aligned}$$

Hence a general solution for the number of mature trees is

$$m = e^{-t/20}\left[(-A-2B)\cos\frac{t}{10} + (2A-B)\sin\frac{t}{10}\right]$$

(d) (i) Use the initial conditions $s(0) = 20$ and $m(0) = 20$.
From the formula for s,

$$s(0) = A = 20$$

so $A = 20$.

From the formula for m,

$$\begin{aligned} m(0) &= -A - 2B = 20 \\ -20 - 2B &= 20 \\ -2B &= 40 \\ B &= -20 \end{aligned}$$

Therefore

$$s = 20e^{-t/20}\left(\cos\frac{t}{10} - \sin\frac{t}{10}\right)$$

To find when no saplings remain, set $s = 0$:

$$20e^{-t/20}\left(\cos\frac{t}{10} - \sin\frac{t}{10}\right) = 0$$

Since $20e^{-t/20} \neq 0$, we need

$$\begin{aligned} \cos\frac{t}{10} - \sin\frac{t}{10} &= 0 \\ \tan\frac{t}{10} &= 1 \end{aligned}$$

The first positive solution is

$$\frac{t}{10} = \frac{\pi}{4}$$

so

$$t = \frac{5\pi}{2} \approx 7.85$$

This is about 7.85 years after the start of 2021, so the year is 2028.

The first value of t is $\frac{5\pi}{2}$ years, so the year is 2028.

(ii) Substituting $A = 20$ and $B = -20$ into the formula for m,

$$m = 20e^{-t/20}\left(\cos\frac{t}{10} + 3\sin\frac{t}{10}\right)$$

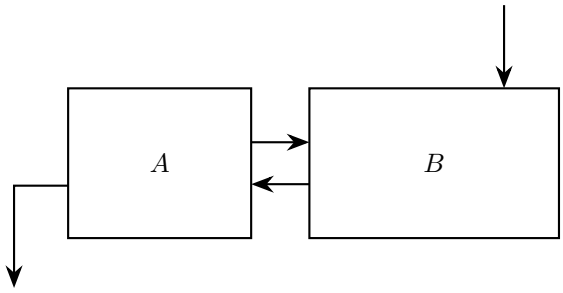
At $t = \frac{5\pi}{2}$,

$$\begin{aligned} m &= 20e^{-(5\pi/2)/20}\left(\cos\frac{\pi}{4} + 3\sin\frac{\pi}{4}\right) \\ &= 20e^{-\pi/8}\left(\frac{1}{\sqrt{2}} + \frac{3}{\sqrt{2}}\right) \\ &= 20e^{-\pi/8}\left(\frac{4}{\sqrt{2}}\right) \\ &= 40\sqrt{2}\,e^{-\pi/8} \\ &\approx 38.2 \end{aligned}$$

So the model predicts about 38.2 mature trees at that time.

(iii) This model is not suitable beyond that time, because after $t = \frac{5\pi}{2}$ it gives $s < 0$, meaning a negative number of saplings, which is impossible.

Also, at $t = \frac{5\pi}{2}$ it predicts about 38.2 mature trees, but numbers of trees should be whole numbers. Therefore the model is only reasonable for a limited time interval and is unsuitable in the longer term.



8. A treatment system consists of two well-stirred tanks, A and B. Tank A has capacity 300 litres and tank B has capacity 500 litres.

At time $t = 0$, tank A is full and contains 300 grams of salt, while tank B is full and contains 1200 grams of salt.

When $t > 0$, liquid flows in the following ways:

- Brine with a salt concentration of μ grams per litre flows into tank B at a constant rate
- Liquid flows from tank A to tank B at a rate of 3 litres per minute
- Liquid flows from tank B to tank A at a rate of r litres per minute
- Liquid flows out of tank A through a waste pipe
- The amount of liquid in tank A remains at 300 litres and the amount of liquid in tank B remains at 500 litres

At time t minutes ($t \geq 0$) there are x grams of salt in tank A and y grams of salt in tank B.

This system is represented by the coupled differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{20}x + \frac{3}{100}y$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{100}x - \frac{3}{100}y + 24$$

(a) Find the value of r **[2]**

(b) Show that $\mu = 2$ **[3]**

(c) Solve the coupled differential equations to find both x and y in terms of t **[9]**

Solution

(a) Since the tanks are well stirred, the salt concentrations are

$$\frac{x}{300}\ \text{g L}^{-1} \quad \text{in tank } A, \qquad \frac{y}{500}\ \text{g L}^{-1} \quad \text{in tank } B$$

Let the waste outflow from tank A be w L min^{-1}. Since the volume of tank A remains constant,

$$r = 3 + w$$

so the total outflow from tank A is r L min^{-1}.

Hence the salt balance for tank A is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{r}{500}y - \frac{r}{300}x$$

Comparing this with

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{20}x + \frac{3}{100}y$$

gives

$$\frac{r}{300} = \frac{1}{20}$$

so

$$r = 15$$

Therefore $r = 15$ litres per minute.

(b) Let the external inflow to tank B be q L min^{-1}. Since the volume of tank B remains constant,

$$q + 3 = r$$

Using $r = 15$ from part (a),

$$q = 15 - 3 = 12$$

Now write the salt balance for tank B.
External inflow contributes salt at rate 12μ g min^{-1}, inflow from A contributes

$$3 \cdot \frac{x}{300} = \frac{x}{100}$$

and outflow from B to A removes salt at rate

$$15 \cdot \frac{y}{500} = \frac{3y}{100}$$

So

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 12\mu + \frac{x}{100} - \frac{3y}{100}$$

Comparing with the given equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 24 + \frac{x}{100} - \frac{3y}{100}$$

we get

$$12\mu = 24$$

and therefore

$$\mu = 2$$

Therefore $\mu = 2$ g L^{-1}.

(c) To solve the system, eliminate y.
From equation (1),

$$\begin{aligned}
\frac{\mathrm{d}x}{\mathrm{d}t} &= -\frac{1}{20}x + \frac{3}{100}y \\
100\frac{\mathrm{d}x}{\mathrm{d}t} &= -5x + 3y \\
3y &= 100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x
\end{aligned}$$

Differentiate equation (1):

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{100}\frac{\mathrm{d}y}{\mathrm{d}t}$$

Now substitute for $\frac{\mathrm{d}y}{\mathrm{d}t}$ from equation (2):

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{100}\left(24 + \frac{1}{100}x - \frac{3}{100}y\right)\\ &= -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{18}{25} + \frac{3}{10000}x - \frac{9}{10000}y\end{aligned}$$

Using

$$y = \frac{100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x}{3}$$

gives

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{18}{25} + \frac{3}{10000}x - \frac{9}{10000}\left(\frac{100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x}{3}\right)\\ &= -\frac{1}{20}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{18}{25} + \frac{3}{10000}x - \frac{3}{100}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2000}x\\ &= -\frac{2}{25}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{3}{2500}x + \frac{18}{25}\end{aligned}$$

Hence

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{2}{25}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{2500}x = \frac{18}{25}$$

We now solve this second order linear equation.
The auxiliary equation is

$$\begin{aligned}\lambda^2 + \frac{2}{25}\lambda + \frac{3}{2500} &= 0\\ (50\lambda + 1)(50\lambda + 3) &= 0\end{aligned}$$

So the roots are

$$\lambda = -\frac{1}{50}, \qquad \lambda = -\frac{3}{50}$$

Therefore the complementary function is

$$x_c = C_1e^{-t/50} + C_2e^{-3t/50}$$

For a particular integral, try a constant $x_p = A$. Then

$$\frac{3}{2500}A = \frac{18}{25}$$

so

$$A = 600$$

Thus

$$x = x_c + x_p = 600 + C_1e^{-t/50} + C_2e^{-3t/50}$$

Now use the initial conditions

$$x(0) = 300, \qquad y(0) = 1200$$

First, from $x(0) = 300$,

$$300 = 600 + C_1 + C_2$$

so

$$C_1 + C_2 = -300$$

Next find $\frac{\mathrm{d}x}{\mathrm{d}t}\big|_{t=0}$ from equation (1):

$$\begin{aligned}\left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} &= -\frac{1}{20}(300) + \frac{3}{100}(1200)\\ &= -15 + 36\\ &= 21\end{aligned}$$

Differentiate the expression for x:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{50}C_1 e^{-t/50} - \frac{3}{50}C_2 e^{-3t/50}$$

Hence at $t = 0$,

$$21 = -\frac{1}{50}C_1 - \frac{3}{50}C_2$$

so

$$1050 = -C_1 - 3C_2$$

Now solve

$$\begin{aligned} C_1 + C_2 &= -300 \\ C_1 + 3C_2 &= -1050 \end{aligned}$$

Subtracting the first equation from the second gives

$$2C_2 = -750$$

so

$$C_2 = -375$$

and then

$$C_1 = 75$$

Therefore

$$x = 600 + 75e^{-t/50} - 375e^{-3t/50}$$

To find y, use

$$3y = 100\frac{\mathrm{d}x}{\mathrm{d}t} + 5x$$

First calculate $\frac{\mathrm{d}x}{\mathrm{d}t}$:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= -\frac{75}{50}e^{-t/50} + \frac{1125}{50}e^{-3t/50} \\ &= -\frac{3}{2}e^{-t/50} + \frac{45}{2}e^{-3t/50} \end{aligned}$$

So

$$100\frac{\mathrm{d}x}{\mathrm{d}t} = -150e^{-t/50} + 2250e^{-3t/50}$$

and

$$5x = 3000 + 375e^{-t/50} - 1875e^{-3t/50}$$

Adding,

$$3y = 3000 + 225e^{-t/50} + 375e^{-3t/50}$$

Hence

$$y = 1000 + 75e^{-t/50} + 125e^{-3t/50}$$

Therefore the solutions are

$$x = 600 + 75e^{-t/50} - 375e^{-3t/50}, \qquad y = 1000 + 75e^{-t/50} + 125e^{-3t/50}$$

9. Two invasive plant species, X and Y, are spreading through a wetland and compete for light and nutrients. A model for this competition assumes, in one particular wetland, the following.

- In the absence of the other species, species X would increase at a rate proportional to the number present, with constant of proportionality p.
- In the absence of the other species, species Y would increase at a rate proportional to the number present, with constant of proportionality q.
- Competition reduces the rate of increase of species X by an amount proportional to the number of species Y present.
- Competition reduces the rate of increase of species Y by an amount proportional to the number of species X present.

If the numbers of species X and Y present at time t years are x and y respectively, the model gives the differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = px - ay \qquad \text{and} \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = qy - bx$$

(a) (i) Show that x satisfies

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - (p+q)\frac{\mathrm{d}x}{\mathrm{d}t} + (pq - ab)x = 0$$

and hence that the general solution for x is

$$x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

where λ_1 and λ_2 are the roots of

$$\lambda^2 - (p+q)\lambda + (pq - ab) = 0$$ **[6]**

(ii) Hence show that the general solution for y can be written in the form

$$y = \frac{p-\lambda_1}{a}Ae^{\lambda_1 t} + \frac{p-\lambda_2}{a}Be^{\lambda_2 t}$$ **[2]**

Observations suggest that suitable values for the model are $p = 0.035$, $q = 0.005$, $a = 0.03$ and $b = 0.0225$. You should use these values in the rest of this question.

(b) When $t = 0$, the numbers present of species X and Y in this wetland are x_0 and y_0 respectively.

(i) Show that

$$x = \frac{1}{4}(3x_0 - 2y_0)e^{0.05t} + \frac{1}{4}(x_0 + 2y_0)e^{-0.01t}$$ **[3]**

(ii) Hence show that

$$y = -\frac{1}{8}(3x_0 - 2y_0)e^{0.05t} + \frac{3}{8}(x_0 + 2y_0)e^{-0.01t}$$ **[1]**

(c) Use initial values $x_0 = 400$ and $y_0 = 500$ with the results in part (b) to determine what the model predicts for each of the following.

(i) What numbers of each species will be present after 25 years? **[2]**

(ii) When will the numbers of the two species be equal? **[4]**

(iii) Does either species ever disappear from the wetland? Justify your answer **[3]**

(d) Different initial values will apply in other wetlands where the two species compete, but previous studies indicate that in most wetlands one population eventually increases while the other decreases.

(i) Find a relation between x_0 and y_0 for which the model does not predict this outcome **[1]**

(ii) Explain what the model predicts in the long term for this exceptional case **[2]**

Solution

(a) (i) Starting from

$$\frac{\mathrm{d}x}{\mathrm{d}t} = px - ay$$

differentiate with respect to t:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = p\frac{\mathrm{d}x}{\mathrm{d}t} - a\frac{\mathrm{d}y}{\mathrm{d}t}$$

Using

$$\frac{\mathrm{d}y}{\mathrm{d}t} = qy - bx$$

gives

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= p\frac{\mathrm{d}x}{\mathrm{d}t} - a(qy - bx)\\ &= p\frac{\mathrm{d}x}{\mathrm{d}t} - aqy + abx\end{aligned}$$

From the first differential equation,

$$ay = px - \frac{\mathrm{d}x}{\mathrm{d}t}$$

so

$$y = \frac{px - \frac{\mathrm{d}x}{\mathrm{d}t}}{a}$$

Substituting this into the expression for $\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$:

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= p\frac{\mathrm{d}x}{\mathrm{d}t} - q\left(px - \frac{\mathrm{d}x}{\mathrm{d}t}\right) + abx\\ &= p\frac{\mathrm{d}x}{\mathrm{d}t} - pqx + q\frac{\mathrm{d}x}{\mathrm{d}t} + abx\\ &= (p+q)\frac{\mathrm{d}x}{\mathrm{d}t} - (pq - ab)x\end{aligned}$$

Rearranging,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - (p+q)\frac{\mathrm{d}x}{\mathrm{d}t} + (pq - ab)x = 0$$

as required.

The auxiliary equation is therefore

$$\lambda^2 - (p+q)\lambda + (pq - ab) = 0$$

If its roots are λ_1 and λ_2, then the general solution is

$$x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

(ii) From

$$\frac{\mathrm{d}x}{\mathrm{d}t} = px - ay$$

we have

$$y = \frac{px - \frac{\mathrm{d}x}{\mathrm{d}t}}{a}$$

Now

$$x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \lambda_1 A e^{\lambda_1 t} + \lambda_2 B e^{\lambda_2 t}$$

Hence

$$\begin{aligned} y &= \frac{p\left(Ae^{\lambda_1 t} + Be^{\lambda_2 t}\right) - \left(\lambda_1 Ae^{\lambda_1 t} + \lambda_2 Be^{\lambda_2 t}\right)}{a} \\ &= \frac{p-\lambda_1}{a} Ae^{\lambda_1 t} + \frac{p-\lambda_2}{a} Be^{\lambda_2 t} \end{aligned}$$

So the general solution for y can be written as

$$y = \frac{p-\lambda_1}{a} Ae^{\lambda_1 t} + \frac{p-\lambda_2}{a} Be^{\lambda_2 t}$$

(b) (i) Using $p = 0.035$, $q = 0.005$, $a = 0.03$ and $b = 0.0225$, the auxiliary equation becomes

$$\begin{aligned} \lambda^2 - (p+q)\lambda + (pq - ab) &= 0 \\ \lambda^2 - 0.04\lambda + (0.035 \times 0.005 - 0.03 \times 0.0225) &= 0 \\ \lambda^2 - 0.04\lambda - 0.0005 &= 0 \end{aligned}$$

Solving,

$$\lambda = \frac{0.04 \pm \sqrt{0.04^2 - 4(1)(-0.0005)}}{2} = \frac{0.04 \pm 0.06}{2}$$

so the roots are

$$\lambda_1 = 0.05, \qquad \lambda_2 = -0.01$$

Therefore

$$x = Ae^{0.05t} + Be^{-0.01t}$$

From part (a)(ii),

$$\begin{aligned} y &= \frac{0.035 - 0.05}{0.03} Ae^{0.05t} + \frac{0.035 - (-0.01)}{0.03} Be^{-0.01t} \\ &= -\frac{1}{2} Ae^{0.05t} + \frac{3}{2} Be^{-0.01t} \end{aligned}$$

Use the initial conditions $x(0) = x_0$ and $y(0) = y_0$:

$$\begin{aligned} A + B &= x_0 \\ -\frac{1}{2}A + \frac{3}{2}B &= y_0 \end{aligned}$$

Multiplying the second equation by 2,

$$-A + 3B = 2y_0$$

From $A = x_0 - B$,

$$\begin{aligned} -(x_0 - B) + 3B &= 2y_0 \\ -x_0 + 4B &= 2y_0 \\ B &= \frac{x_0 + 2y_0}{4} \end{aligned}$$

Then

$$\begin{aligned} A &= x_0 - \frac{x_0 + 2y_0}{4} \\ &= \frac{4x_0 - x_0 - 2y_0}{4} \\ &= \frac{3x_0 - 2y_0}{4} \end{aligned}$$

Substituting into the expression for x,

$$x = \frac{1}{4}(3x_0 - 2y_0)e^{0.05t} + \frac{1}{4}(x_0 + 2y_0)e^{-0.01t}$$

(ii) Substituting these values of A and B into

$$y = -\frac{1}{2}Ae^{0.05t} + \frac{3}{2}Be^{-0.01t}$$

gives

$$\begin{aligned} y &= -\frac{1}{2}\left(\frac{3x_0 - 2y_0}{4}\right)e^{0.05t} + \frac{3}{2}\left(\frac{x_0 + 2y_0}{4}\right)e^{-0.01t} \\ &= -\frac{1}{8}(3x_0 - 2y_0)e^{0.05t} + \frac{3}{8}(x_0 + 2y_0)e^{-0.01t} \end{aligned}$$

Hence

$$y = -\frac{1}{8}(3x_0 - 2y_0)e^{0.05t} + \frac{3}{8}(x_0 + 2y_0)e^{-0.01t}$$

(c) (i) With $x_0 = 400$ and $y_0 = 500$,

$$3x_0 - 2y_0 = 1200 - 1000 = 200, \qquad x_0 + 2y_0 = 400 + 1000 = 1400$$

So

$$\begin{aligned} x &= \frac{1}{4}(200)e^{0.05t} + \frac{1}{4}(1400)e^{-0.01t} \\ &= 50e^{0.05t} + 350e^{-0.01t} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{1}{8}(200)e^{0.05t} + \frac{3}{8}(1400)e^{-0.01t} \\ &= -25e^{0.05t} + 525e^{-0.01t} \end{aligned}$$

At $t = 25$,

$$\begin{aligned} x(25) &= 50e^{1.25} + 350e^{-0.25} \approx 447.1 \\ y(25) &= -25e^{1.25} + 525e^{-0.25} \approx 321.4 \end{aligned}$$

So after 25 years the model predicts about 447 of species X and about 321 of species Y.

(ii) Set $x = y$:

$$\begin{aligned} 50e^{0.05t} + 350e^{-0.01t} &= -25e^{0.05t} + 525e^{-0.01t} \\ 75e^{0.05t} &= 175e^{-0.01t} \end{aligned}$$

Divide by $75e^{-0.01t}$:

$$e^{0.06t} = \frac{175}{75} = \frac{7}{3}$$

Taking natural logarithms,

$$\begin{aligned} 0.06t &= \ln\left(\frac{7}{3}\right) \\ t &= \frac{\ln(7/3)}{0.06} \approx 14.1 \end{aligned}$$

So the numbers of the two species are equal after about 14.1 years.

(iii) For species X,

$$x = 50e^{0.05t} + 350e^{-0.01t}$$

Both exponentials are positive for all $t \geq 0$, and both coefficients are positive, so

$$x > 0 \quad \text{for all } t \geq 0$$

Therefore species X never disappears.

For species Y, set $y = 0$:

$$\begin{aligned} -25e^{0.05t} + 525e^{-0.01t} &= 0 \\ 25e^{0.05t} &= 525e^{-0.01t} \\ e^{0.06t} &= 21 \end{aligned}$$

Hence

$$t = \frac{\ln 21}{0.06} \approx 50.7$$

So the model predicts that species Y disappears after about 50.7 years.

(d) (i) From part (b),

$$x = \frac{1}{4}(3x_0 - 2y_0)e^{0.05t} + \frac{1}{4}(x_0 + 2y_0)e^{-0.01t}$$

and

$$y = -\frac{1}{8}(3x_0 - 2y_0)e^{0.05t} + \frac{3}{8}(x_0 + 2y_0)e^{-0.01t}$$

In the long term, the $e^{0.05t}$ term dominates unless its coefficient is zero. So the exceptional case is

$$3x_0 - 2y_0 = 0$$

that is

$$3x_0 = 2y_0$$

(ii) If $3x_0 = 2y_0$, then the growing term vanishes and

$$\begin{aligned} x &= \frac{1}{4}(x_0 + 2y_0)e^{-0.01t} \\ y &= \frac{3}{8}(x_0 + 2y_0)e^{-0.01t} \end{aligned}$$

Since $y_0 = \frac{3}{2}x_0$,

$$\begin{aligned} x &= \frac{1}{4}(x_0 + 3x_0)e^{-0.01t} = x_0e^{-0.01t} \\ y &= \frac{3}{8}(x_0 + 3x_0)e^{-0.01t} = \frac{3}{2}x_0e^{-0.01t} = y_0e^{-0.01t} \end{aligned}$$

Therefore both populations decrease exponentially to 0 as $t \to \infty$, with ratio

$$y : x = 3 : 2$$

So in this exceptional case neither species eventually increases; both die away.

10. A pharmacologist is studying the movement of a drug between a patient's bloodstream and muscle tissue. The amount of drug in the bloodstream after t hours is x mg. The amount of drug in the muscle tissue after t hours is y mg. Initially, when $t = 0$, there is no drug in either the bloodstream or the muscle tissue.

At first it is assumed that no drug is removed from the body. The pharmacologist further assumes that the drug is infused into the bloodstream at a constant rate of q mg per hour, where $q > 6$.

For $t \geq 0$, this situation is modelled by the following pair of first-order linear differential equations.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -5x + 30y + q$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = 5x - 30y$$

(a) (i) Show that

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 35\frac{\mathrm{d}x}{\mathrm{d}t} = 30q$$ **[2]**

(ii) Hence determine x in terms of q and t **[7]**

(iii) The patient's bloodstream is tested 20 hours after the infusion begins. Explain why, according to this model, the patient will be at risk of toxic side effects when the test is carried out **[1]**

Further observations suggest that some of the drug is metabolised in the muscle tissue and so is removed from the body. The model is refined in light of this. One resulting expression for x is

$$x = q\left(10 - e^{-17t}\left(10\cosh\left(\sqrt{241}\,t\right) + \frac{169}{\sqrt{241}}\sinh\left(\sqrt{241}\,t\right)\right)\right)$$

In the refined model, it is still assumed that the drug is infused at a constant rate.

(b) Given now that $q < 9$, determine whether the patient will be at risk of toxic side effects in the long run according to this refined model **[2]**

(c) Suggest a reason why it might be more realistic to model the infusion rate as not being constant **[1]**

Solution

(a) (i) Differentiate the first equation with respect to t:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -5\frac{\mathrm{d}x}{\mathrm{d}t} + 30\frac{\mathrm{d}y}{\mathrm{d}t}$$

From

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -5x + 30y + q$$

we rearrange to get

$$5x - 30y = q - \frac{\mathrm{d}x}{\mathrm{d}t}$$

Using the second equation,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 5x - 30y = q - \frac{\mathrm{d}x}{\mathrm{d}t}$$

Substitute this into the differentiated equation:

$$\begin{aligned}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} &= -5\frac{\mathrm{d}x}{\mathrm{d}t} + 30\left(q - \frac{\mathrm{d}x}{\mathrm{d}t}\right)\\ &= 30q - 35\frac{\mathrm{d}x}{\mathrm{d}t}\end{aligned}$$

Hence

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 35\frac{\mathrm{d}x}{\mathrm{d}t} = 30q$$

This is the required result.

(ii) Initially, $x = 0$ and $y = 0$ when $t = 0$. So from the first differential equation,

$$\frac{\mathrm{d}x}{\mathrm{d}t}(0) = -5(0) + 30(0) + q = q$$

Let

$$u = \frac{\mathrm{d}x}{\mathrm{d}t}$$

Then the equation from part (i) becomes

$$\frac{\mathrm{d}u}{\mathrm{d}t} + 35u = 30q$$

with

$$u(0) = q$$

Use the integrating factor e^{35t}:

$$\begin{aligned}e^{35t}\frac{\mathrm{d}u}{\mathrm{d}t} + 35e^{35t}u &= 30qe^{35t}\\ \frac{\mathrm{d}}{\mathrm{d}t}\left(ue^{35t}\right) &= 30qe^{35t}\end{aligned}$$

Integrating,

$$\begin{aligned}ue^{35t} &= \int 30qe^{35t}\,\mathrm{d}t\\ &= \frac{30q}{35}e^{35t} + C\\ &= \frac{6q}{7}e^{35t} + C\end{aligned}$$

Therefore

$$u = \frac{6q}{7} + Ce^{-35t}$$

Use $u(0) = q$:

$$\begin{aligned}q &= \frac{6q}{7} + C\\ C &= \frac{q}{7}\end{aligned}$$

So

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{6q}{7} + \frac{q}{7}e^{-35t}$$

Now integrate to find x:

$$\begin{aligned}x &= \int\left(\frac{6q}{7} + \frac{q}{7}e^{-35t}\right)\mathrm{d}t\\ &= \frac{6q}{7}t - \frac{q}{245}e^{-35t} + C\end{aligned}$$

Use $x(0) = 0$:

$$\begin{aligned} 0 &= 0 - \frac{q}{245} + C \\ C &= \frac{q}{245} \end{aligned}$$

Hence

$$\begin{aligned} x &= \frac{6q}{7}t - \frac{q}{245}e^{-35t} + \frac{q}{245} \\ &= q\left(\frac{6t}{7} + \frac{1 - e^{-35t}}{245}\right) \end{aligned}$$

Therefore

$$x = q\left(\frac{6t}{7} + \frac{1 - e^{-35t}}{245}\right)$$

(iii) At $t = 20$,

$$x(20) = q\left(\frac{120}{7} + \frac{1 - e^{-700}}{245}\right)$$

Since $q > 6$ and $\dfrac{1 - e^{-700}}{245} > 0$,

$$x(20) > 6 \cdot \frac{120}{7} \approx 102.86 > 100$$

So according to this model, the patient will be at risk of toxic side effects when the test is carried out.

(b) In the long run, we consider $t \to \infty$.

Using

$$\cosh z = \frac{e^z + e^{-z}}{2} \qquad \sinh z = \frac{e^z - e^{-z}}{2}$$

we have

$$\begin{aligned} e^{-17t}\cosh(\sqrt{241}\,t) &= \frac{1}{2}\left(e^{-(17-\sqrt{241})t} + e^{-(17+\sqrt{241})t}\right) \\ &\to 0 \end{aligned}$$

and

$$\begin{aligned} e^{-17t}\sinh(\sqrt{241}\,t) &= \frac{1}{2}\left(e^{-(17-\sqrt{241})t} - e^{-(17+\sqrt{241})t}\right) \\ &\to 0 \end{aligned}$$

because $17 > \sqrt{241}$.

Therefore, from

$$x = q\left(10 - e^{-17t}\left(10\cosh\left(\sqrt{241}\,t\right) + \frac{169}{\sqrt{241}}\sinh\left(\sqrt{241}\,t\right)\right)\right)$$

it follows that

$$x \to 10q \qquad \text{as } t \to \infty$$

Given $q < 9$,

$$10q < 90 < 100$$

So according to the refined model, the patient will not be at risk of toxic side effects in the long run.

(c) A constant infusion rate may be unrealistic because, in practice, the rate may be adjusted over time according to the patient's condition or to keep the drug level within a safe range.

11. A water-treatment plant uses two storage tanks, tank A and tank B, joined by a balancing pipe. At any given instant, water can be transferred between the two tanks. It is assumed that these transfers happen instantaneously.

The volume of water in tank A at time t hours is denoted by A and the corresponding volume in tank B is B. Because the volumes are large for most of the time, A and B are modelled as being continuous.

When the volumes in the two tanks are different, water is pumped from the tank with more water to the tank with less water. In addition, water flows directly into tank A at a rate proportional to t^2. An engineer therefore decides to model A and B with the following coupled differential equations.

$$\frac{\mathrm{d}A}{\mathrm{d}t} = k(B - A) + \gamma t^2$$
$$\frac{\mathrm{d}B}{\mathrm{d}t} = k(A - B)$$

(a) Explain why k must be positive. **[1]**

After examining data, the engineer chooses $k = 1$ and $\gamma = 4$.

(b) Show that A satisfies the second order differential equation

$$\frac{\mathrm{d}^2A}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A}{\mathrm{d}t} = 4t^2 + 8t$$

[2]

(c) (i) Find the complementary function for the differential equation from part (b). **[2]**

(ii) Explain why a particular integral of the form $A = at^2 + bt + c$ will not work in this situation. **[1]**

(iii) Using a particular integral of the form $A = at^3 + bt^2 + ct$, find the general solution of the differential equation from part (b). **[3]**

(d) At a certain time, tank A contains $52\,\mathrm{m}^3$ of water and the volume in tank A is increasing at $13\,\mathrm{m}^3\,\mathrm{h}^{-1}$. Use the model, starting from this time, to estimate the volume of water in tank A 30 minutes later. **[4]**

(e) Explain why the model becomes unreliable as t gets very large. **[1]**

Solution

(a) If $B > A$, water should be transferred from tank B to tank A, so the term $k(B - A)$ in

$$\frac{\mathrm{d}A}{\mathrm{d}t} = k(B - A) + \gamma t^2$$

must be positive. Since $B - A > 0$, this requires $k > 0$.

Similarly, if $A > B$, then $B - A < 0$, so $k(B - A) < 0$, which makes A decrease as expected. Hence k must be positive.

(b) With $k = 1$ and $\gamma = 4$,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = B - A + 4t^2, \qquad \frac{\mathrm{d}B}{\mathrm{d}t} = A - B$$

Differentiate the first equation:

$$\frac{\mathrm{d}^2A}{\mathrm{d}t^2} = \frac{\mathrm{d}B}{\mathrm{d}t} - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t$$

Now substitute $\frac{\mathrm{d}B}{\mathrm{d}t} = A - B$:

$$\frac{\mathrm{d}^2 A}{\mathrm{d}t^2} = A - B - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t$$

From

$$\frac{\mathrm{d}A}{\mathrm{d}t} = B - A + 4t^2$$

we have

$$B - A = \frac{\mathrm{d}A}{\mathrm{d}t} - 4t^2$$

so

$$A - B = -\frac{\mathrm{d}A}{\mathrm{d}t} + 4t^2$$

Therefore

$$\begin{aligned}\frac{\mathrm{d}^2 A}{\mathrm{d}t^2} &= \left(-\frac{\mathrm{d}A}{\mathrm{d}t} + 4t^2\right) - \frac{\mathrm{d}A}{\mathrm{d}t} + 8t \\ &= 4t^2 + 8t - 2\frac{\mathrm{d}A}{\mathrm{d}t}\end{aligned}$$

Hence

$$\frac{\mathrm{d}^2 A}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A}{\mathrm{d}t} = 4t^2 + 8t$$

So the required equation is shown.

(c) (i) The auxiliary equation is

$$\lambda^2 + 2\lambda = 0$$

Factorising,

$$\lambda(\lambda + 2) = 0$$

so the roots are $\lambda = 0$ and $\lambda = -2$. Therefore

$$A_c = C_1 + C_2 e^{-2t}$$

So the complementary function is $A_c = C_1 + C_2 e^{-2t}$.

(ii) If

$$A_p = at^2 + bt + c$$

then

$$\frac{\mathrm{d}A_p}{\mathrm{d}t} = 2at + b, \qquad \frac{\mathrm{d}^2 A_p}{\mathrm{d}t^2} = 2a$$

Hence

$$\frac{\mathrm{d}^2 A_p}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A_p}{\mathrm{d}t} = 2a + 2(2at + b) = 4at + (2a + 2b)$$

This is only linear in t, but the right-hand side is $4t^2 + 8t$, which contains a t^2 term. Therefore a particular integral of the form $at^2 + bt + c$ will not work.

(iii) Take

$$A_p = at^3 + bt^2 + ct$$

Then

$$\frac{\mathrm{d}A_p}{\mathrm{d}t} = 3at^2 + 2bt + c, \qquad \frac{\mathrm{d}^2 A_p}{\mathrm{d}t^2} = 6at + 2b$$

Substitute into

$$\frac{\mathrm{d}^2 A}{\mathrm{d}t^2} + 2\frac{\mathrm{d}A}{\mathrm{d}t} = 4t^2 + 8t$$

to get

$$\begin{aligned}(6at + 2b) + 2(3at^2 + 2bt + c) &= 4t^2 + 8t \\ 6at^2 + (6a + 4b)t + (2b + 2c) &= 4t^2 + 8t\end{aligned}$$

Equating coefficients of powers of t,

$$6a = 4, \qquad 6a + 4b = 8, \qquad 2b + 2c = 0$$

So

$$a = \frac{2}{3}, \qquad b = 1, \qquad c = -1$$

Thus

$$A_p = \frac{2}{3}t^3 + t^2 - t$$

Combining with the complementary function,

$$A = C_1 + C_2 e^{-2t} + \frac{2}{3}t^3 + t^2 - t$$

So the general solution is

$$A = C_1 + C_2 e^{-2t} + \frac{2}{3}t^3 + t^2 - t$$

(d) Take the stated instant as $t = 0$. Then

$$A(0) = 52, \qquad \left.\frac{\mathrm{d}A}{\mathrm{d}t}\right|_{t=0} = 13$$

From part (c),

$$A = C_1 + C_2 e^{-2t} + \frac{2}{3}t^3 + t^2 - t$$

Differentiate:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -2C_2 e^{-2t} + 2t^2 + 2t - 1$$

Use $\left.\frac{\mathrm{d}A}{\mathrm{d}t}\right|_{t=0} = 13$:

$$-2C_2 - 1 = 13$$

so

$$-2C_2 = 14$$

and hence

$$C_2 = -7$$

Now use $A(0) = 52$:

$$C_1 + C_2 = 52$$

so

$$C_1 - 7 = 52$$

giving

$$C_1 = 59$$

Therefore

$$A(t) = 59 - 7e^{-2t} + \frac{2}{3}t^3 + t^2 - t$$

After 30 minutes, $t = 0.5$. So

$$\begin{aligned} A(0.5) &= 59 - 7e^{-1} + \frac{2}{3}(0.5)^3 + (0.5)^2 - 0.5 \\ &= 59 - \frac{7}{e} + \frac{1}{12} + \frac{1}{4} - \frac{1}{2} \\ &= 59 - \frac{7}{e} - \frac{1}{6} \end{aligned}$$

Hence

$$A(0.5) \approx 56.3$$

So the estimated volume of water in tank A 30 minutes later is 56.3 m^3.

(e) As t gets very large, the inflow term $4t^2$ becomes arbitrarily large, so the model predicts the volume will keep increasing without bound. In reality the tanks have finite capacity and operating conditions would not remain the same, so the model becomes unreliable for large t.